

1. Network Protocol Fundamentals

- Overview of network layers and protocols
- Common vulnerabilities in protocol implementations

2. Network-Based Attacks

- TCP attacks (SYN flood, TCP reset, session hijacking)
- ARP spoofing and poisoning
- Packet sniffing and spoofing techniques

3. Vulnerability Analysis

- MITRE ATT&CK and D3FEND frameworks
- Using MITRE ATT&CK Navigator for threat modeling
- Understanding CVE (Common Vulnerabilities and Exposures)
- CWE (Common Weakness Enumeration), NVD (National Vulnerability Database), and CVSS (Common Vulnerability Scoring System)

4. Security Infrastructure Components

- Firewalls: Types, configurations, and policies
- Intrusion Detection Systems (IDS): Signature-based and anomaly-based detection

5. Attack Tools: Network Scanning

- Reconnaissance and enumeration techniques
- Tools for scanning open ports, services, and vulnerabilities

6. Security Testing and Analysis Tools

- Wireshark for protocol-level traffic inspection
- Burp Suite for web application security testing
- Metasploitable for exploit simulation and penetration testing

7. Risk Analysis and Threat Modeling

- Introduction to risk assessment methodologies
- Performing structured risk analysis using YACRAF framework